Answer Key

1. B
2. C
3. C
4. A
5. C
6. C
7. C
8. B
9. C
10. D
11. C
12. A
13. D
14. C
15. B
16. A
17. A
18. C
19. C
20. B
21. D
22. C
23. C
24. D
25. B
26. A
27. B
28. C
29. B
30. B

Detailed Solutions

1. $$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
   $$P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7.$$
   Answer: B.

2. $$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$
   $$P(A \mid B) = \frac{0.18}{0.3} = 0.6.$$
   Answer: C.

3. Exactly one of A and B occurs if A occurs without B, or B occurs without A:
   $$P(\text{exactly one}) = P(A) + P(B) - 2P(A \cap B)$$
   $$= 0.45 + 0.35 - 2(0.10) = 0.60.$$
   Answer: C.

4. Check independence:
   $$P(A)P(B) = 0.4(0.5) = 0.2.$$
   Since
   $$P(A \cap B) = 0.2,$$
   we have
   $$P(A \cap B) = P(A)P(B).$$
   Therefore, A and B are independent. Answer: A.

5. The sum of two dice is even when both dice have the same parity: both odd or both even.
   $$P(\text{both odd}) = \frac{3}{6} \cdot \frac{3}{6} = \frac{1}{4}$$
   $$P(\text{both even}) = \frac{3}{6} \cdot \frac{3}{6} = \frac{1}{4}$$
   Thus
   $$P(\text{sum even}) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$
   Answer: C.

6. The maximum is at least 5 unless both rolls are at most 4.
   $$P(\max \geq 5) = 1 - P(\max \leq 4)$$
   $$= 1 - \left(\frac{4}{6}\right)^2 = 1 - \frac{16}{36} = \frac{20}{36} = \frac{5}{9}.$$
   Answer: C.

7. For a binomial random variable,
$$E[X] = np.$$
Thus
$$E[X] = 20(0.4) = 8.$$
Answer: C.

8. For a binomial random variable,
$$\text{Var}(X) = np(1-p).$$
Thus
$$\text{Var}(X) = 20(0.4)(0.6) = 4.8.$$
Answer: B.

9. For $X \sim \text{Binomial}(5, 0.2)$,
$$P(X=0) = \binom{5}{0}(0.2)^0(0.8)^5.$$
Since
$$\binom{5}{0} = 1$$
and
$$(0.2)^0 = 1,$$
we get
$$P(X=0) = (0.8)^5.$$
Answer: C.

10. For a Poisson random variable,
$$E[X] = \lambda$$
and
$$\text{Var}(X) = \lambda.$$
Also,
$$\text{Var}(X) = E[X^2] - (E[X])^2.$$
Therefore,
$$E[X^2] = \text{Var}(X) + (E[X])^2.$$
Since $\lambda = 3$,
$$E[X^2] = 3 + 3^2 = 3 + 9 = 12.$$
Answer: D.

11. For $X \sim \text{Poisson}(2)$,
$$P(X \le 1) = P(X=0) + P(X=1).$$
Using the Poisson formula,
$$P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!},$$
we get
$$P(X=0) = e^{-2}$$

and

$$P(X = 1) = 2e^{-2}.$$

Thus

$$P(X \leq 1) = e^{-2} + 2e^{-2} = 3e^{-2}.$$

Answer: C.

12. Since $p(x)$ is a pmf, the probabilities must sum to 1:

$$\sum_x p(x) = 1.$$

So

$$c(0+1) + c(1+1) + c(2+1) = 1.$$
$$c + 2c + 3c = 1.$$
$$6c = 1.$$

Thus

$$c = \frac{1}{6}.$$

Answer: A.

13. Using $c = 1/6$,

$$P(X \geq 1) = P(X = 1) + P(X = 2).$$
$$P(X = 1) = \frac{1}{6}(2) = \frac{2}{6}$$

and

$$P(X = 2) = \frac{1}{6}(3) = \frac{3}{6}.$$

Thus

$$P(X \geq 1) = \frac{2}{6} + \frac{3}{6} = \frac{5}{6}.$$

Answer: D.

14. Using the pmf,

$$E[X] = \sum_x xp(x).$$

The probabilities are

$$P(X = 0) = \frac{1}{6}, \quad P(X = 1) = \frac{2}{6}, \quad P(X = 2) = \frac{3}{6}.$$

Thus

$$E[X] = 0\left(\frac{1}{6}\right) + 1\left(\frac{2}{6}\right) + 2\left(\frac{3}{6}\right).$$
$$E[X] = \frac{2}{6} + \frac{6}{6} = \frac{8}{6} = \frac{4}{3}.$$

Answer: C.

15. First compute $E[X^2]$:

$$E[X^2] = \sum_x x^2 p(x).$$

$$E[X^2] = 0^2\left(\frac{1}{6}\right) + 1^2\left(\frac{2}{6}\right) + 2^2\left(\frac{3}{6}\right).$$

$$E[X^2] = \frac{2}{6} + \frac{12}{6} = \frac{14}{6} = \frac{7}{3}.$$

Then

$$\text{Var}(X) = E[X^2] - (E[X])^2.$$

$$\text{Var}(X) = \frac{7}{3} - \left(\frac{4}{3}\right)^2 = \frac{7}{3} - \frac{16}{9} = \frac{21}{9} - \frac{16}{9} = \frac{5}{9}.$$

Answer: B.

16. Since $f(x)$ is a pdf,

$$\int_0^4 cx\,dx = 1.$$

$$c\int_0^4 x\,dx = 1.$$

$$c\left[\frac{x^2}{2}\right]_0^4 = 1.$$

$$c\left(\frac{16}{2}\right) = 1.$$

$$8c = 1.$$

Thus

$$c = \frac{1}{8}.$$

Answer: A.

17. Using $c = 1/8$,

$$P(X \le 2) = \int_0^2 \frac{1}{8}x\,dx.$$

$$P(X \le 2) = \frac{1}{8}\left[\frac{x^2}{2}\right]_0^2 = \frac{1}{8}\cdot\frac{4}{2} = \frac{1}{8}\cdot 2 = \frac{1}{4}.$$

Answer: A.

18.

$$E[X] = \int_0^4 x f(x)\,dx.$$

Since

$$f(x) = \frac{1}{8}x,$$

we have

$$E[X] = \int_0^4 x\left(\frac{1}{8}x\right)\,dx = \frac{1}{8}\int_0^4 x^2\,dx.$$

$$E[X] = \frac{1}{8}\left[\frac{x^3}{3}\right]_0^4 = \frac{1}{8} \cdot \frac{64}{3} = \frac{8}{3}.$$

Answer: C.

19.
$$P(X > 2) = 1 - P(X \leq 2).$$

Using the cdf,
$$F(2) = \frac{(2-1)^2}{4} = \frac{1}{4}.$$

Thus
$$P(X > 2) = 1 - \frac{1}{4} = \frac{3}{4}.$$

Answer: C.

20. Since
$$X \sim N(40, 16),$$

the mean is 40 and the standard deviation is
$$\sigma = \sqrt{16} = 4.$$

Compute the z-score:
$$z = \frac{36 - 40}{4} = -1.$$

Thus
$$P(X < 36) = P(Z < -1) \approx 0.159.$$

Answer: B.

21. Since
$$X \sim N(40, 16),$$

we have $\sigma = 4$. Compute both z-scores:
$$z_1 = \frac{32 - 40}{4} = -2,$$

and
$$z_2 = \frac{48 - 40}{4} = 2.$$

Thus
$$P(32 < X < 48) = P(-2 < Z < 2).$$

Using the standard normal rule,
$$P(-2 < Z < 2) \approx 0.954.$$

Answer: D.

22. Since

$$X \sim N(100, 25),$$

the mean is 100 and the standard deviation is

$$\sigma = \sqrt{25} = 5.$$

We need

$$P(X \leq x) = 0.841.$$

For the standard normal distribution,

$$P(Z \leq 1) \approx 0.841.$$

So the corresponding z-score is approximately 1:

$$z = \frac{x - 100}{5} = 1.$$

Solving,

$$x - 100 = 5,$$

so

$$x = 105.$$

Answer: C.

23.

$$P(X = 1) = P(X = 1, Y = 0) + P(X = 1, Y = 1).$$

From the table,

$$P(X = 1) = 0.1 + 0.4 = 0.5.$$

Answer: C.

24.

$$P(Y = 1 \mid X = 1) = \frac{P(X = 1, Y = 1)}{P(X = 1)}.$$

From the previous problem,

$$P(X = 1) = 0.5.$$

Also,

$$P(X = 1, Y = 1) = 0.4.$$

Thus

$$P(Y = 1 \mid X = 1) = \frac{0.4}{0.5} = 0.8.$$

Answer: D.

25. First compute

$$E[X] = P(X = 1) = 0.5.$$

Also,

$$E[Y] = P(Y = 1) = P(X = 0, Y = 1) + P(X = 1, Y = 1).$$

$$E[Y] = 0.3 + 0.4 = 0.7.$$

Next,

$$E[XY] = \sum_x \sum_y xyP(X = x, Y = y).$$

Since $XY = 1$ only when $X = 1$ and $Y = 1$,

$$E[XY] = 1 \cdot 1 \cdot P(X = 1, Y = 1) = 0.4.$$

Therefore,

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

$$\mathrm{Cov}(X, Y) = 0.4 - (0.5)(0.7) = 0.4 - 0.35 = 0.05.$$

Answer: B.

26. The sample mean has mean

$$E[\bar{X}] = \mu = 20.$$

The standard deviation of $\bar{X}$ is

$$\frac{\sigma}{\sqrt{n}} = \frac{5}{\sqrt{100}} = \frac{5}{10} = 0.5.$$

Compute the z-score:

$$z = \frac{21 - 20}{0.5} = 2.$$

Thus

$$P(\bar{X} > 21) = P(Z > 2) \approx 0.023.$$

Answer: A.

27. The sample mean has mean

$$E[\bar{X}] = \mu = 10.$$

The standard deviation of $\bar{X}$ is

$$\frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{64}} = \frac{4}{8} = 0.5.$$

Compute the z-score:

$$z = \frac{9.5 - 10}{0.5} = -1.$$

Thus

$$P(\bar{X} < 9.5) = P(Z < -1) \approx 0.159.$$

Answer: B.

28. Chebyshev's inequality says

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

Here,

$$\mu = 50, \qquad \sigma^2 = 16, \qquad k = 8.$$

Thus

$$P(|X - 50| \geq 8) \leq \frac{16}{8^2} = \frac{16}{64} = 0.25.$$

Answer: C.

29. We want

$$P(6 < X < 18).$$

Since the mean is 12, this is the same as

$$P(|X - 12| < 6).$$

Chebyshev's inequality gives

$$P(|X - 12| \geq 6) \leq \frac{9}{6^2} = \frac{9}{36} = 0.25.$$

Therefore,

$$P(|X - 12| < 6) \geq 1 - 0.25 = 0.75.$$

Answer: B.

30. First compute the covariance.

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Since

$$\sigma_X = \sqrt{9} = 3, \qquad \sigma_Y = \sqrt{16} = 4,$$

we have

$$0.5 = \frac{\mathrm{Cov}(X, Y)}{12}.$$

Thus

$$\mathrm{Cov}(X, Y) = 6.$$

Now use

$$\mathrm{Var}(aX + bY) = a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y) + 2ab\, \mathrm{Cov}(X, Y).$$

With $a = 2$ and $b = -3$,

$$\mathrm{Var}(2X - 3Y) = (2)^2(9) + (-3)^2(16) + 2(2)(-3)(6).$$

$$= 36 + 144 - 72.$$

$$= 108.$$

Answer: B.